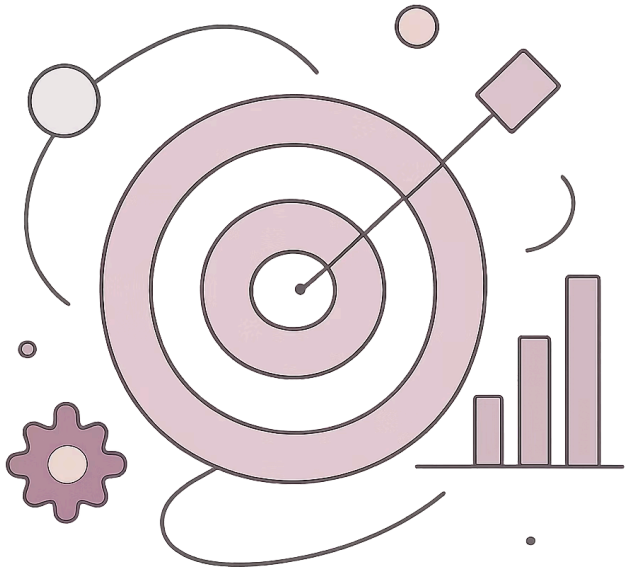




Ad Campaign A/B Testing

A data-driven analysis comparing **Facebook** and **Instagram** advertising performance for XYZ Company's digital product — using Python, statistical testing, and visual analytics.

Project Overview



Objective

Determine which advertising platform — Facebook or Instagram — is most effective at driving revenue for XYZ Company's digital product.

Approach

Full exploratory data analysis, feature engineering, visualization, and statistical hypothesis testing using Python.

Dataset at a Glance

365

Rows

Daily records across one full year

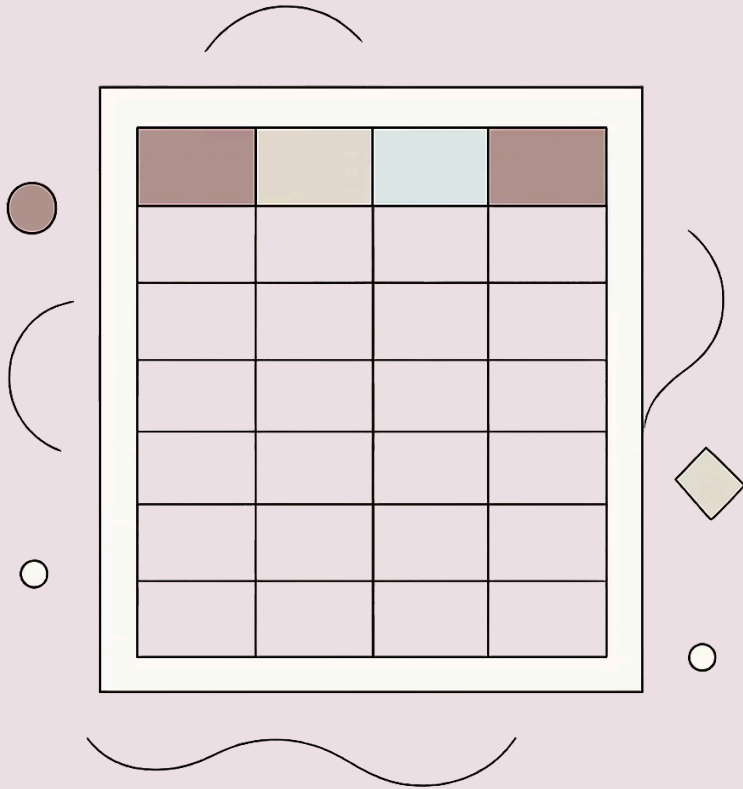
9

Columns

Key advertising metrics

Key Features

- Date
- Facebook: Ad Views, Clicks, Conversions, Cost/Day
- Instagram: Ad Views, Clicks, Conversions, Cost/Day



Exploratory Data Analysis

Data preparation and cleaning in Python using pandas.

01

Data Loading

Imported dataset via pandas; used `df.info()` and `.describe()` for structure and summary statistics.

03

Column Standardization

Renamed columns to snake_case for readability and documentation.

02

Missing Data Check

Checked for null values — none found.

04

Feature Engineering

Created **Facebook Conversion Category** and **Instagram Conversion Category** columns based on daily conversion counts.

Correlation Analysis

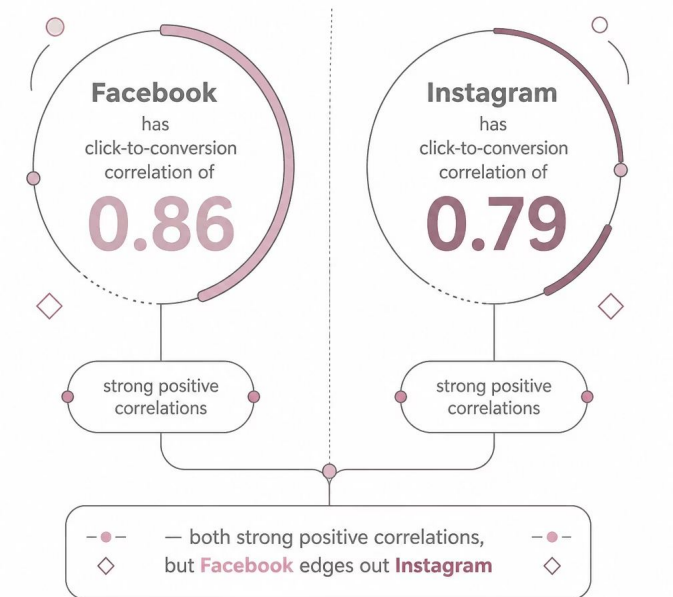
Facebook: Clicks vs. Conversions

Correlation: 0.86 — a very strong positive association between clicks and conversions.

Instagram: Clicks vs. Conversions

Correlation: 0.79 — also a strong positive association, but slightly weaker than Facebook.

❏ **Early finding:** Facebook outperforms Instagram in click-to-conversion strength.



Distribution Analysis

Histograms were created to examine the distribution of clicks and conversions for both campaigns and identify visible patterns.

Facebook Distribution

Plotted to assess conversion frequency and spread across the 365-day period.

Instagram Distribution

Plotted for direct comparison against Facebook's conversion patterns.

Observation

Visual inspection alone was inconclusive — statistical testing was needed for a definitive conclusion.



Conversion Categories

Daily conversions were grouped into four tiers to enable category-level comparison between platforms.

< 200

Low conversion days

200 – 400

Moderate conversion days

400 – 600

High conversion days

> 600

Very high conversion days

Both Facebook and Instagram daily conversions were categorized, counted, and merged for visual side-by-side comparison.

Visual Comparison: Facebook vs. Instagram



Key Observation

Visual comparison of conversion categories reveals that **Facebook consistently generates more conversions** across higher-tier categories than Instagram.

- ✓ Facebook outperforms Instagram across conversion volume categories – reinforcing the correlation findings.

T-Test: Statistical Validation

Hypotheses

H₀: Facebook and Instagram average conversions are equal.

H₁: Facebook and Instagram average conversions are different.

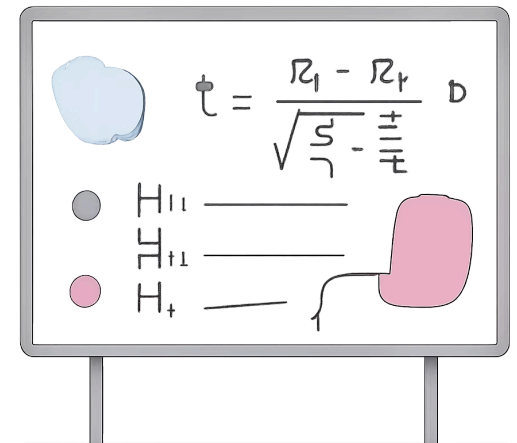
Results

t-statistic = 58.94

Extremely large and positive — Facebook mean is much higher.

p-value = 1.28×10^{-198}

Far below significance level of 0.05.



Conclusions

Facebook Mean

788.53 average daily conversions

Instagram Mean

134.28 average daily conversions

Verdict

H_0 rejected. Difference is **statistically significant** — Welch's t-test: $t = 58.94$, $p = 1.28 \times 10^{-198}$

- ✔ **Facebook outperforms Instagram by a large margin** across all analyses — correlation, distribution, and statistical testing confirm the difference is genuine and not due to chance.

